

Giri's Tech Hub Pvt.Ltd, Pune.

Programming (Machine) Test

Batch: June-24 to June-25

Date: 29/05/2026
Time: 10:00 to 12:00 Pm

Instructions:

1. Solve any 10 query.

1. Create Table Course :-

Field	Type	Null	Key	Default	Extra
CourseID	INT	NO	PRI	NULL	AUTO_INCREMENT
CourseName	VARCHAR(100)	NO		NULL	
Duration	INT	NO		NULL	

2. Create Table Batches :-

Field	Type	Null	Key	Default	Extra
BatchID	INT	NO	PRI	NULL	AUTO_INCREMENT
BatchName	VARCHAR(100)	NO		NULL	
StartDate	DATE	NO		NULL	
CourseID	INT	NO	MUL	NULL	

3. Create Table Instructors :-

Field	Type	Null	Key	Default	Extra
InstructorID	INT	NO	PRI	NULL	AUTO_INCREMENT
Name	VARCHAR(100)	NO		NULL	
CourseID	INT	NO	MUL	NULL	

4. Create Table Students :-

Field	Type	Null	Key	Default	Extra
StudentID	INT	NO	PRI	NULL	AUTO_INCREMENT
Name	VARCHAR(100)	NO		NULL	
Age	INT	NO		NULL	CHECK (Age >= 18)
CourseID	INT	YES	MUL	NULL	
BatchID	INT	YES	MUL	NULL	

1. List all batches along with their corresponding course names.

2. Count the number of students enrolled in each course.

- 3. Retrieve all students along with their batch names and course names.**
- 4. Find the batch with the maximum number of students enrolled.**
- 5. Find the instructor name, course name, and the number of students enrolled in each instructor's course.**
- 6. List all students, their course, batch name, and instructor name (if assigned).**
- 7. Find students who are older than the average age of all students.**
- 8. Find the students who have the same instructor.**
- 9. Retrieve the courses with at least 5 students enrolled.**
- 10. Create a view named StudentsSameCourseDiffBatchView that displays students who are in the same course but different batches.**
- 11. Create a view named StudentsWithoutInstructorView that displays students who are enrolled in a course but have no assigned instructor.**
- 12. Write a trigger that prevents adding a student to a batch if the batch already has 10 students.**
- 13. Write a SQL function to calculate the sum of all even numbers up to a given number using a loop.**
- 14. Write a stored procedure to check if a number is Armstrong.**
 - The procedure should take an integer as input and return TRUE if the number is armstrong, otherwise return FALSE.**
- 15. Write a SQL procedure to get the total number of students enrolled in each course.**

-----ALL THE BEST-----